Insider threat, backdoor, Logic bombs, Malware and it’s attack

- Insider threat
- Backdoor/trapdoor of a computer system
- Logic bombs
- Computer viruses, its kind and prevention
- Trojan horse & its types
- Computer worm
- Rootkits
- Bot, its types and botnets
- Privacy invasive software
- Bacteria/rabbit
- Countermeasures against the malware
- Economics of malware

Insider threat

What is an insider threat?

- An insider threat is a security risk that occurs when a current or former employee or business associate misuses their access to an organization's network.
- An insider threat can involve leaking data, sabotaging equipment, or stealing company property. Insider threats can be difficult to detect and combat.
- An insider threat is a security risk that comes from within your company. Employees, partners, vendors, interns, suppliers or contractors can potentially become an insider threat. These people can access your organization’s internal network and may accidentally leak or purposely steal sensitive information.

Types of insider threats

Types of insider threats

Here, 6 types of insider threats and the risky data movements are mentioned:

1. **Departing employees:** Employees leaving the company voluntarily or involuntarily are among the most common insider threats. They might take materials they're proud of to help land a new job or, more viciously, steal and expose sensitive data out of revenge.
2. **Malicious insiders:** These individuals are current employees. They might not be your company's biggest fans and usually act on their grievances by altering or deleting crucial data sets, disclosing secret information or engaging in other types of sabotage.
3. **Negligent workers:** Although insider threat management strategies often focus on malicious insiders, careless workers are more dangerous. These employees can unintentionally put organizations at risk by not applying proper security hygiene like strong passwords, multi-factor authentication, or allowing others to use their work device.

Types of insider threats

4. **Security evaders:** Modern companies have security policies for safeguarding their essential data. Some workers can find these protections inconvenient, leading them to create **workarounds** that increase the chances of a data breach.
5. **Inside agents:** These threats work on behalf of an external group, whether knowingly or unknowingly. Outsiders may compel them to give information through blackmail or bribery or deceive them into sharing their login credentials through social engineering.
6. **Third-party partners:** Not all insiders are on the payroll. Suppliers, contractors, vendors and other external parties with some level of inside access can be just as dangerous as employees with the same permissions.

Knowing how insider threats manifest can help you safeguard your company from them, protecting your organization's reputation, future success, customers and employees.

Defences against insider attacks

Defences against insider attacks

1. **Educate employees**

   Provide training and education on data handling policies to help employees understand their role in protecting company resources and the limits of their data access.

2. **Implement access controls**

   Use access controls and policies to limit who can access sensitive data and reduce the risk of unauthorized access.

3. **Apply the principle of least privilege**

   Only give users access to the specific data, resources, and applications they need to perform their job.

4. **Conduct risk assessments**

   Regularly assess your systems to ensure they are secure and up to date with the latest security measures.

Defences against insider attacks

5. Use threat modelling

Use threat modelling to identify and prevent cyber attacks, including those from insiders.

6. Monitor user behaviour

Use tools to monitor user behavior and detect anomalies.

7. Establish incident response procedures

Have procedures in place for responding to and reporting incidents.

8. Identify sensitive assets

Assess how accessible sensitive assets are to different insiders and identify potential threats.

Backdoor /trapdoor of a computer system

Backdoor/trapdoor of a computer system

- A backdoor/trapdoor is a method that allows unauthorized users to bypass security mechanisms and gain access to a system. Backdoors can be created for legitimate purposes, but they can also be used by attackers to gain access to a system.
- A backdoor **threat** is a cyber attack that gives hackers unauthorized access to a system, network, or application. Backdoors can be created by design flaws, coding errors, or malicious code.
- They can remain undetected for weeks, and even when vulnerabilities are identified, it can be difficult to determine where they end.
- Once an attacker has access to a system through a backdoor, they can potentially modify files, steal personal information, install unwanted software, and even take control of the entire computer or computer system.

Dangers of backdoor threats

- Stealing personal information
- Installing unwanted software
- Taking control of the entire computer
- Changing the infrastructure
- Making the system behave as the attacker wants
- Launching distributed denial of service (DDoS) attacks
- Infecting website visitors

Logic bombs

Logic bombs

- A logic bomb is a type of malicious code that is hidden in a computer system or application and executes when certain conditions are met. The code remains dormant until the trigger conditions are satisfied, and then it can cause a range of harmful outcomes.

- Logic bombs can be used for a variety of reasons, including revenge, profiteering, or just to cause havoc. For example, a disgruntled employee or spy might insert a logic bomb into a company system that deletes files when they are terminated.

- Some common triggers for logic bombs include: A specific date or time, Event-based triggers, User actions, and System conditions.

Range of destructive actions of logic bombs

- Deleting files
- Corrupting data
- Crashing systems
- Disrupting critical systems
- Data loss
- Financial losses
- Loss of productivity

Logic bombs are different from computer viruses and worms because they do not self-replicate or spread to other systems.

Malware

- Malware, or malicious software, is a type of software that is designed to disrupt a computer or network, steal data, or damage a computer system.

OR

- Malware refers to any **intrusive** software developed by cybercriminals (often called hackers) to steal data and damage or destroy computers and computer systems.

- The goal of malware is to cause harm, either for financial gain or sabotage.

- Examples of common malware include **viruses, worms, Trojan, rootkit spyware, adware, and ransomware.**

Computer viruses

Computer viruses

- A computer virus is a type of malicious software, or malware, that spreads between computers and causes damage to data and software.

- Computer viruses aim to disrupt systems, cause major operational issues, and result in data loss and leakage. A key thing to know about computer viruses is that they are designed to spread across programs and systems.

- Computer viruses typically **attach to an executable host file**, which results in their viral codes executing when a file is opened. The code then spreads from the document or software it is attached to **via networks, drives, file-sharing programs, or infected email attachments**.

How Do Computer Viruses Attack and Spread?

- In the early days of computers, viruses were spread between devices using floppy disks. Nowadays, viruses can still be spread via hard disks and Universal Serial Bus (USB) devices, but they are more likely to be passed between devices through the internet.

- Computer viruses can be spread via email, with some even capable of hijacking email software to spread themselves

- **A key feature of any computer virus is it requires a victim to execute its code or payload**, which means the host application should be running.

Phases of a computer virus

Phases of a computer virus

1. **Dormant phase**
   - The virus enters the computer or software but doesn't do anything.
2. **Propagation phase**
   - The virus multiplies and inserts copies of itself into other programs or areas on the disk.
3. **Triggering phase**
   - The virus changes from dormant to activated by **a specific user action or predefined date.**
4. **Execution/Action phase**
   - The virus releases its payload, which is the malicious code that harms the device.

Types of Computer Viruses

Types of Computer Viruses

1. Resident virus & non-resident virus

- A resident virus is a type of computer malware that embeds itself in the computer's memory (RAM), allowing it to remain active and infect other files even after the original infected program is closed.
- **A resident virus achieves this by infecting applications as they are opened by a user**. A non-resident virus is capable of infecting executable files when programs are not running.

2. Multipartite virus

- **It will typically remain in the computer's memory to infect the hard disk, then spread through and infect more drives by altering the content of applications.** This results in performance lag and application memory running low.

Types of Computer Viruses

3. Direct action

- **A direct action virus accesses a computer's main memory and infects all programs, files, and folders located in the autoexec.bat path, before deleting itself**. This virus typically alters the performance of a system but is capable of destroying all data on the computer's hard disk and any USB device attached to it.

4. Browser hijacker

- **A browser hijacker changes the settings of web browsers, such as replacing the homepage, editing the new tab page, and changing the default search engine without user permission.** Technically, it is not a virus because it cannot infect files but can be hugely damaging to computer users, who often will not be able to restore their homepage or search engine.

Types of Computer Viruses

5. Overwrite virus

- Overwrite viruses are extremely dangerous. **They can delete data and replace it with their own file content or code.** Once files get infected, they cannot be replaced, and the virus can affect Windows, DOS, Linux, and Apple systems. **The only way this virus can be removed is by deleting all of the files it has infected, which could be devastating.**

6. Web scripting virus

- **A web scripting virus attacks web browser security, enabling a hacker to inject web-pages with malicious code, or client-side scripting.** This allows cyber criminals to attack major websites, such as social networking sites, email providers, and any site that enables user input or reviews. Attackers can use the virus to send spam, commit fraudulent activity, and damage server files.

Types of Computer Viruses

7. File infector

- **It overwrites files when they are opened and can quickly spread across systems and networks. It largely affects files with .exe or .com extensions.** The best way to avoid file infector viruses is to only download official software and deploy an antivirus solution.

8. Network Virus

- Network viruses are extremely dangerous because they can completely cripple entire computer networks. They are often difficult to discover, as **the virus could be hidden within any computer on an infected network**. These viruses can easily replicate and spread by using the internet to transfer to devices connected to the network.

Types of Computer Viruses

9. Boot Sector Virus

- **A boot sector virus targets a computer's master boot record (MBR). The virus injects its code into a hard disk's partition table, then moves into the main memory when a computer restarts.**

- The presence of the virus is signified by boot-up problems, poor system performance, and the hard disk becoming unable to locate. Most modern computers come with boot sector safeguards that restrict the potential of this type of virus.

Encrypted viruses

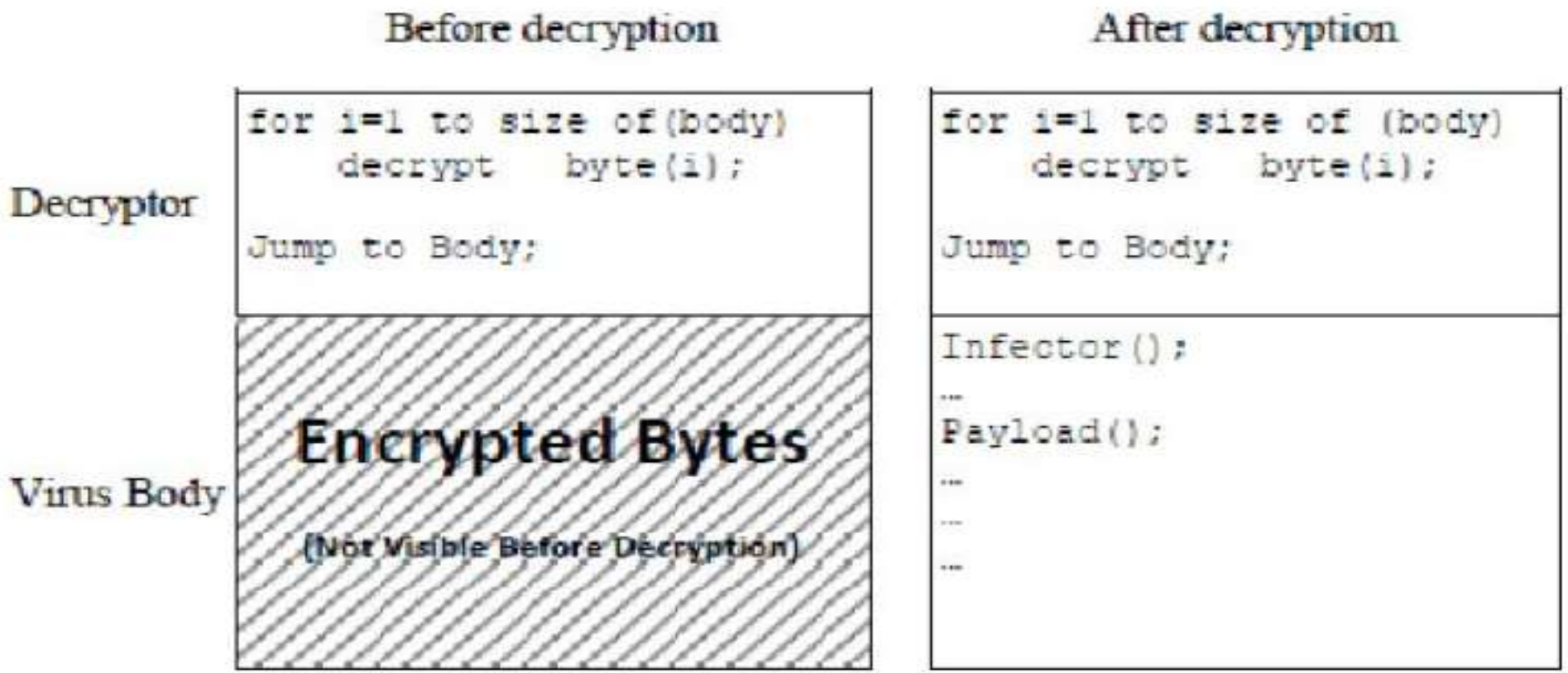
Encrypted viruses

- **An encrypted virus is a type of malware that uses encryption to conceal its code and avoid detection by security software.**

- This technique makes the virus appear as gibberish (*i.e.* unintelligible, meaningless), preventing traditional signature-based antivirus programs from recognizing and stopping it.

- The simplest form, where the malware consists of an encrypted virus body and a small, unencrypted "**decryptor**" program. The decryptor contains the key and routine to unscramble the main virus body

Encrypted viruses



Types of Encrypted viruses

(Polymorphic and metamorphic viruses)

Polymorphic and metamorphic viruses

Polymorphic (genetic variation) and metamorphic (change its code to avoid detection) viruses are both types of malware that can change their code to evade detection and remain on a system.

Polymorphic

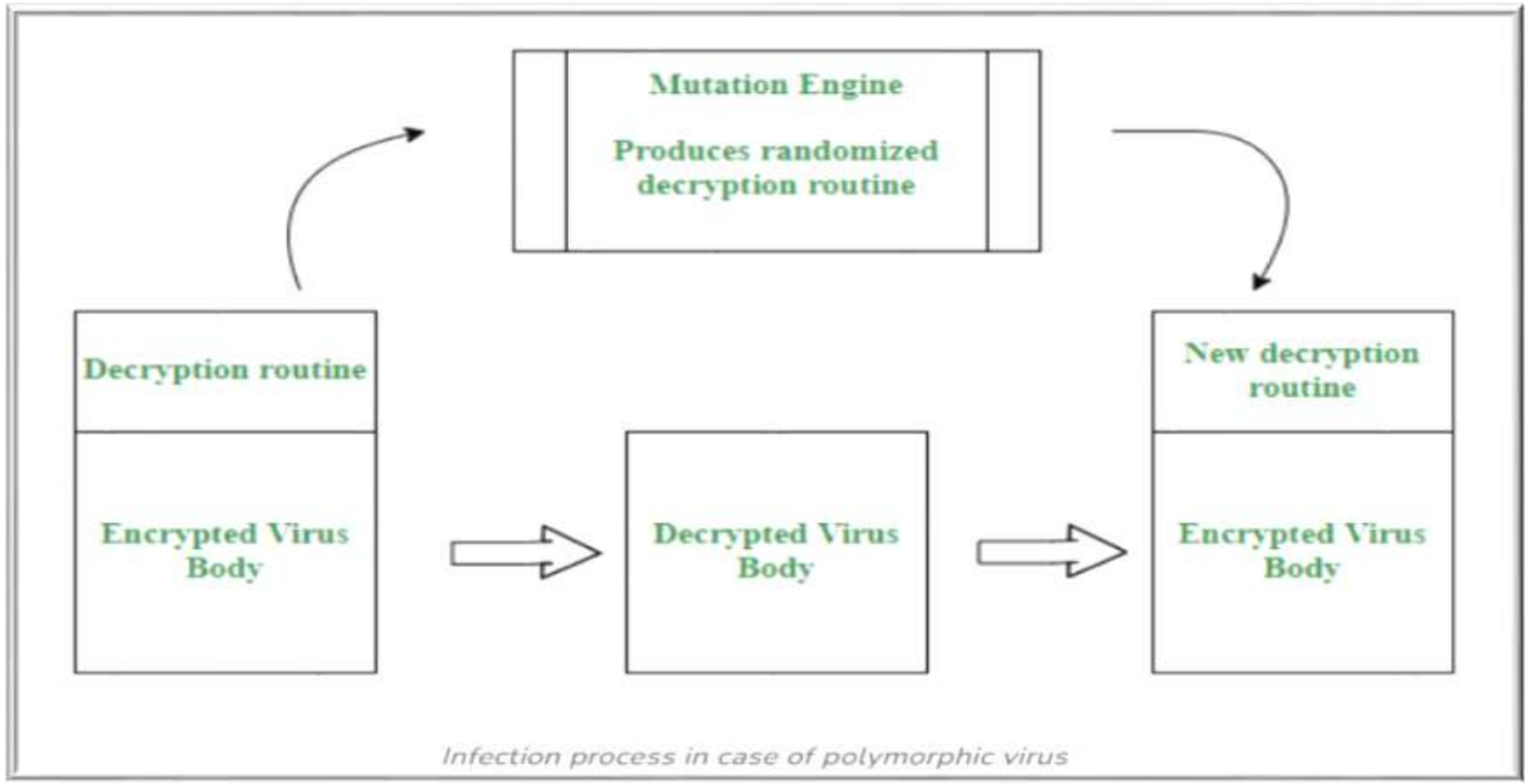
- **Uses a variable encryption key to change its code and make each iteration appear different.** This makes it difficult for traditional antivirus software to recognize and block the threat.

Metamorphic

- **Rewrites its code without an encryption key.** This makes it more complex and transformative than polymorphic malware, and it can be difficult for antivirus scanners to detect.

Polymorphic malware is more common than metamorphic malware. While both types of malware are a threat to systems, polymorphic malware is easier to identify than metamorphic malware because only part of its code is changed.

Working of Polymorphic Virus



In the case of the polymorphic virus, with each mutation, the basic function performed by the virus remains the same even though the signature or decryption routine changes

Detection of Polymorphic Virus

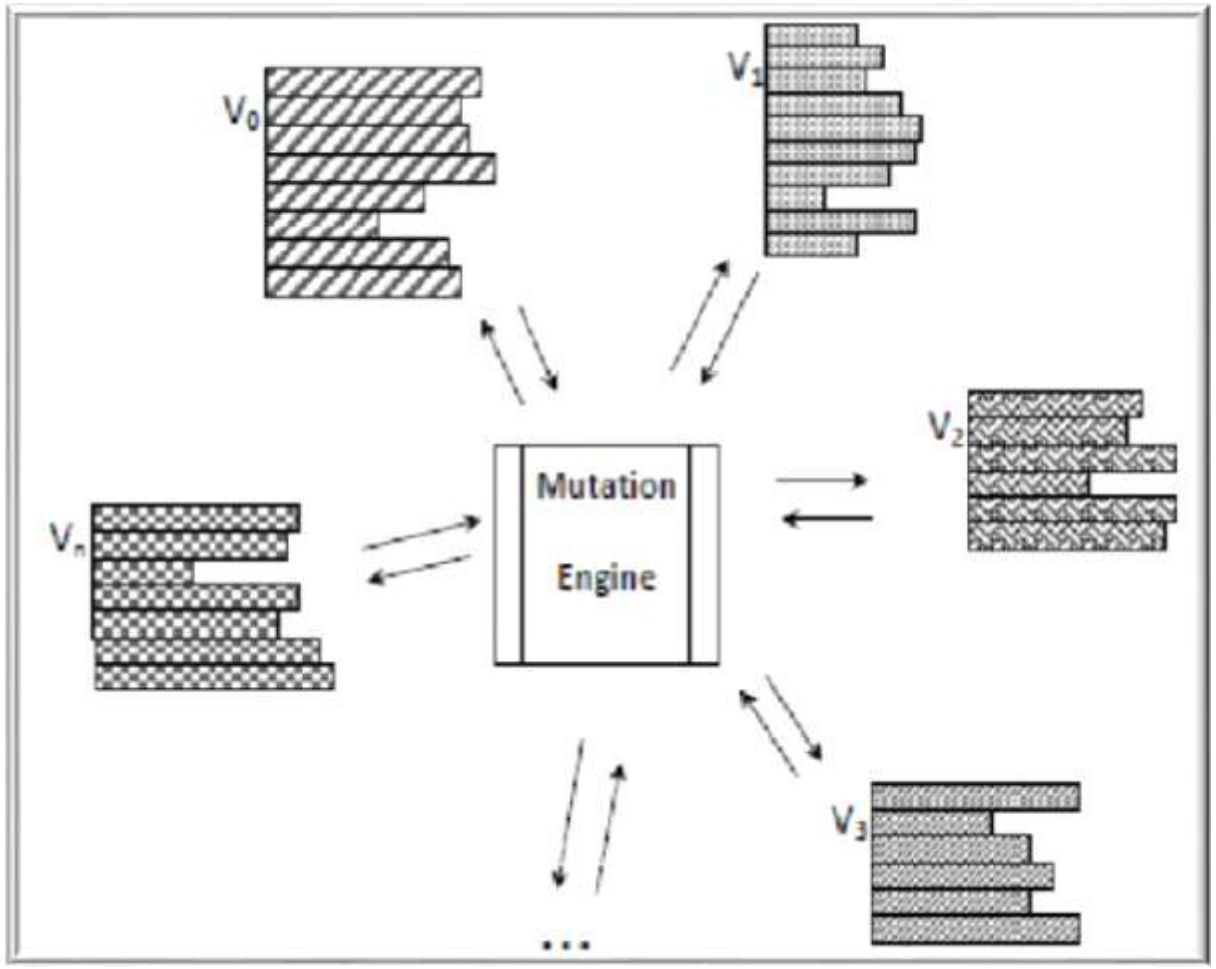
Polymorphic viruses can easily fool the conventional antivirus software using signature-based detection. However, these viruses can be detected by new security technologies which use machine learning and behaviour detection to identify any anomalous behaviour in the system.

- **Behaviour-based detection:** This technique analyses not only the code but also the behaviour of the virus. This helps in detecting viruses with similar behaviour.

- **Heuristic Scanning:** This technique looks for the components that different threats share instead of looking for an exact match to a threat. This helps in detecting new variation viruses.

Preventing/defenses Computer From Viruses

Metamorphic viruses



- Metamorphic malware have similar functionality, different structure and signature.

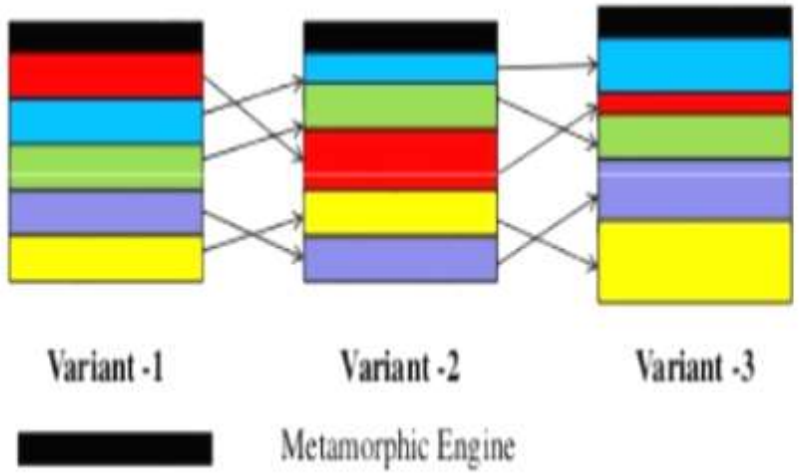


Diagram depicts metamorphic malware variants with reordered code

- Similar to genetic diversity in Biology.

To change code structure at each infection metamorphic malware employee the Mutation Engine (black-part) that use obfuscation techniques like: garbage code insertion, register/variable usage exchange, instruction replacement, instruction permutation, code transposition etc.

Differences between polymorphic and metamorphic viruses

Feature	Polymorphic Virus	Metamorphic Virus
Core Technique	Encrypts its body and uses a variable decryptor.	Rewrites its entire code, including the engine.
Signature	The encrypted body changes, but the decryptor routine often has a detectable signature (though it's hidden with junk code).	No part remains the same. There is no constant decryptor or body.
Analogy	A letter locked in a safe. The safe (encryption) and the lock (decryptor) change, but the letter inside (the malicious code) is the same.	The entire letter is rewritten in a different language and handwriting, but it says the same thing.
Complexity	Less complex.	Extremely complex to create.

Virus signature

Virus signature

- **A virus signature is a unique pattern of code or data that identifies a specific computer virus, similar to a fingerprint.**

- Antivirus software scans files, comparing them against a database of known signatures.

- If a match is found, the software can then take action to remove or isolate the threat.

Limitations of virus signatures

1. **Only detects known threats:**

- A traditional signature-based scanner can only identify malware for which a signature has already been created. This leaves a vulnerability period, known as a "zero-day," between a new malware's release and the antivirus database's update.

2. Vulnerable to polymorphic viruses:

- Sophisticated malware, known as polymorphic or metamorphic viruses, can alter its code with each new infection. This creates a new signature every time, making it difficult for traditional signature-based methods to keep up.

Preventing Computer From Viruses

Preventing Computer From Viruses

1. Use a trusted antivirus product

- Trusted computer antivirus products are crucial to **stop malware attacks** and prevent computers from being infected with viruses. These **antivirus concepts** will protect devices from being infected through regular scans and identifying and blocking malware.

2. Avoid clicking pop-up advertisements

- Unwanted pop-up advertisements are more than likely to be linked to computer viruses and malware. Never click on pop-up advertisements because this can lead to inadvertently downloading viruses onto a computer.

3. Scanning of email attachments

- A popular way to protect your device from computer viruses is to avoid suspicious email attachments, which are commonly used to spread malware. Computer antivirus solutions can be used to scan email attachments for potential viruses.

4. Scan the files that downloads using file-sharing programs

- File-sharing programs, particularly unofficial sites, are also popular resources for attackers to spread computer viruses. Avoid downloading applications, games, or software from unofficial sites, and always scan files that have been downloaded from any file-sharing program.



Trojan horse

Trojan horse

- **A Trojan Horse is a type of malware that downloads onto a computer disguised as a legitimate program.**

OR

- **A Trojan horse is a malicious program that appears to perform a desired function but actually has hidden programs that can steal information, create backdoors, or run scripts.**

- The delivery method typically sees an attacker use social engineering to hide malicious code within legitimate software to try and gain users' system access with their software.

- Trojans can be installed through a backdoor and are often confused with viruses. However, Trojans are usually stand-alone programs and don't inject themselves into other programs.

- Malicious code for a Trojan horse can be injected into benign applications, hidden in JavaScript pages, or masqueraded in email links.

Trojan Horse: malicious actions

Once installed, the Trojan Horse can perform a variety of malicious actions, including:

- Stealing sensitive data, such as passwords and financial information
- Corrupting or deleting files
- Taking control of the device for malicious purposes
- Installing other malware
- Disrupting the device's operation.

Types of Trojan horses

- **Backdoor Trojan**: Includes a backdoor that gives the attacker remote administrative control over the system
- **DDoS Trojan**: Starts Denial of Service (DoS) attacks by sending many requests to the target address
- **Trojan Downloader**: Downloads and installs files, often malicious programs
- **Info-stealer Trojan**: Also known as the information thief, this type of Trojan steals important data from susceptible systems
- **Banker Trojan**: Targets a user's financial account information and attempts to gain access to bank, investment, and credit card accounts
- **Zeus**: Also known as Zbot, this Trojan was designed to steal information such as credit card details and passwords from computers
- **CryptoLocker**: Encrypts files on your device and any connected media

Computer worms

Computer worms

- A computer worm is a type of malware that can replicate itself and spread to other computers without human interaction.
- Worms can exploit security vulnerabilities in a computer to gain access, and then use that computer as a host to infect other computers.
- A computer worm duplicates itself to spread to uninfected computers.
- As it spreads, a worm consumes bandwidth, overloading infected systems and making them unreliable or unavailable.
- Worms can also change and delete files or introduce other malware.

Attack by computer worm

Worms can:

- **Steal information**: Worms can exploit vulnerabilities in security software to steal sensitive information.
- **Install backdoors**: Worms can install backdoors that hackers can use to access a system.
- **Corrupt files**: Worms can corrupt files.
- **Crash systems**: Worms can crash systems through self-replication.
- **Download malicious applications**: Worms can download malicious applications.
- **Overload systems**: Worms can consume large amounts of memory and bandwidth, which can overload and malfunction systems, servers, and networks.

Worms can be difficult to eradicate because they spread quickly and automatically. Even if a worm doesn't have a malicious payload, it can still be a nuisance for IT managers.

Difference between virus and worm

- The main difference between viruses and worms is the method in which they reproduce and spread (how they spread and are activated).

- A virus is dependent upon a host file or boot sector, and the transfer of files between machines to spread, while a worm can run completely independently and spread of its own will through network connections.

- Worms can travel across networks and replicate themselves on a system, potentially consuming a lot of system memory or network bandwidth. Whereas the viruses need to be activated by a user's interaction with an infected file or program.

Computer Rootkits

Computer Rootkits

- A computer rootkit is a type of malware that gives cybercriminals unauthorized access to a computer or network. Rootkits are designed to be difficult to detect and can remain hidden for a long time.

- Rootkit is a type of malware program that enables cyber criminals to gain access to and infiltrate data from machines without being detected.

- It covers software toolboxes designed to infect computers, give the attacker remote control, and remain hidden for a long period of time.

A rootkit is a type of malware that provides a hacker with complete control over a computer, while a Trojan is a type of malware that gives an attacker limited access to a system.

Is A rootkit a Worm? Detection of rootkit!

- Rootkit and worms are two different type of malwares but we might see a blended malware.
- **For example, a worm might spread across network or devices and it contains payload for rootkit or it might contains dropper and it drop a Trojan or Virus which will execute a Rootkit.**
- **A dropper is a type of malicious software that installs other malware onto a device, allowing criminals to bypass security measures and deploy harmful programs. Droppers are typically used during the first stage of a malware attack.**
- A rootkit is a dangerous malware designed to gain **“illegal'' and persistent access to what is otherwise not allowed**. Typically, a rootkit also masks its existence or the existence of other malware.

- **Because the infected programs still run normally, rootkit detection is difficult for users – but antivirus programs can detect them since they both operate on the application layer.**

Features of dropper

1. **Disguising themselves**
   - Droppers can be disguised as harmless files, such as software installers, or as genuine apps with functionalities like calculators or QR code scanners.
2. **Exploiting vulnerabilities**
   - Droppers can exploit vulnerabilities in document formats like PDFs or Microsoft Office files.
3. **Installing payloads**
   - Droppers can install other malicious tools, called payloads, on the target device without the victim noticing.
4. **Neutralizing system defences**
   - Droppers can neutralize system defences before installing their payload.

Feature	Rootkit	Worm
Primary Purpose	To hide itself and other malware; to maintain persistent, undetected access to a system.	To self-replicate and spread to other computers, consuming resources and causing disruption.
Main Behavior	Stealth & Concealment. Modifies the OS core to hide files, processes, and network connections.	Propagation & Spread. Copies itself across networks, email, or vulnerabilities.
Method of Spread	Does not self-replicate. It is typically installed by another threat (e.g., a Trojan, worm, or hacker).	Self-replicating. It spreads automatically without user intervention, often by exploiting vulnerabilities.
User Interaction	Often requires a separate component to be installed. The user may not be directly involved in its installation.	Can spread with no user interaction, or may use social engineering (e.g., enticing email attachments).
Primary Target	The local system's integrity and security. It aims to control that specific machine.	Any and all vulnerable systems it can reach on a network. Its goal is widespread infection.
Key Danger	Stealthy Control. An attacker gains long-term, "root" (administrator) level access to your system.	Network Congestion & Resource Drain. Can overload networks and systems, causing slowdowns or crashes.
Visibility	Designed to be invisible to the user and standard antivirus software.	Its activity (network traffic, slow performance) is often more noticeable, even if the file is hidden.
Analogy	A sleeper agent who lives in your house, hiding in a secret room and monitoring your every move.	A contagious virus that spreads rapidly from person to person, causing a widespread epidemic.
Removal	Very difficult, as it subverts the tools used to detect it. Often requires specialized anti-rootkit software or an OS reinstall.	Can be removed by standard antivirus software, but its rapid spread makes containment a priority.

Bacteria/Rabbit

- Viruses that perform **no explicit malicious behaviour** are often called **bacteria** or **rabbits**. Unlike typical destructive malware (e.g., those that delete files, steal data, or encrypt systems), these viruses simply replicate themselves continuously.
- Even though they don't directly damage data or steal information, they can still cause **serious negative impacts** on a computer system and network.
- **Goal/Effect:** Consume system resources (CPU, memory, disk space) until the system slows down, crashes, or becomes unusable.
- **Spreading Method:** Usually confined to a single system (local replication).
- **Example:**
  - The **Rabbit Virus (1974)**
  - **Fork Bombs** in Unix/Linux.
- **Impact:** Resource exhaustion → Denial of Service (DoS).

Comparison of Rabbit/bacteria & worm

Feature	Bacteria/Rabbits	Traditional Viruses	Worms
Replication	Local, exponential self-replication	Attach to host files, replicate when executed	Network-based, self-propagating
Malicious Payload	None (just replication)	Often destructive (delete/steal data)	May carry destructive payload
Resource Consumption	Very high (crashes system)	Moderate (depends on payload)	High (spreads rapidly over networks)
Spreading Method	Within one machine	Via infected files/media	Across networks automatically
Impact Type	Denial of Service (DoS)	Data loss, corruption, theft	Network slowdown, mass infection, ransomware

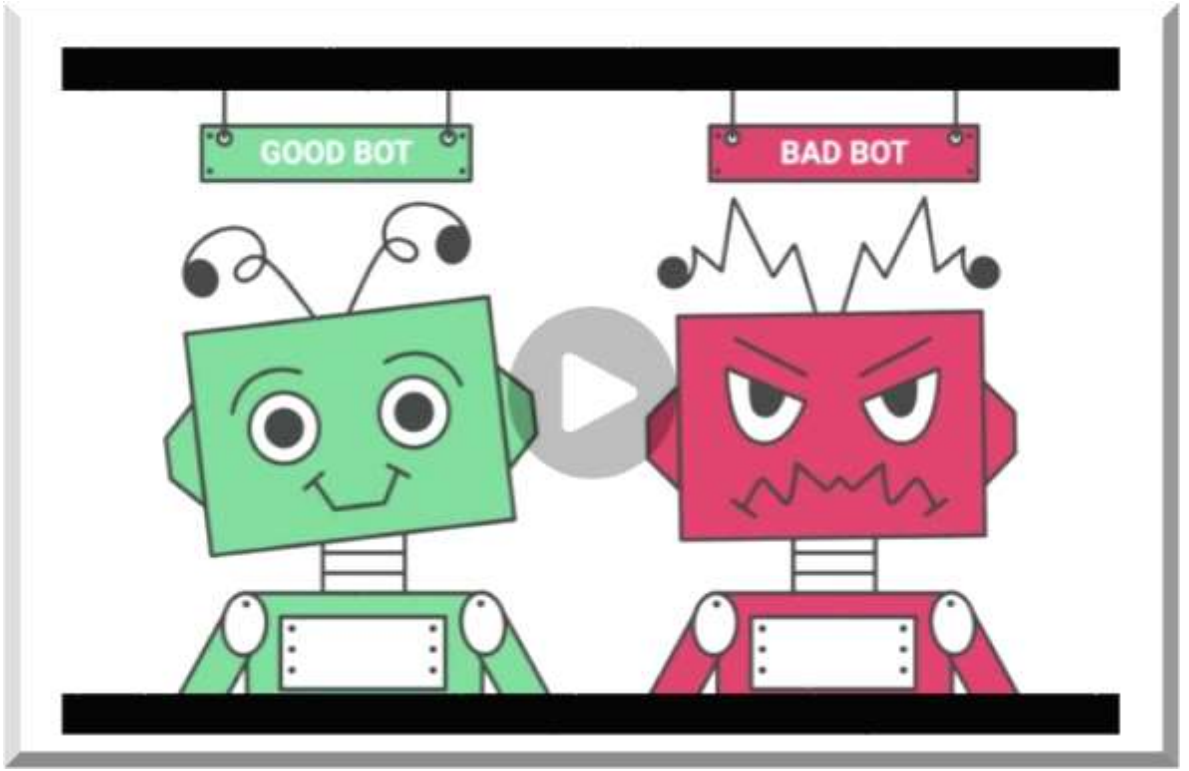
Key historical examples of malware

- **Bacteria/rabbits**: Harmless-looking but harmful due to **resource exhaustion**.
- **Viruses** : Destructive or manipulative, tied to host files.
- **Worms** : Fast-spreading across networks, often with dangerous payloads.
- **1974 – Rabbit Virus** → Endless replication, crashed systems (DoS).
- **1988 – Morris Worm** → First major Internet worm, overloaded thousands of systems.
- **2000 – ILOVEYOU Virus** → Email-based, overwrote files, spread globally.
- **2017 – WannaCry Worm** → Ransomware worm exploiting Windows SMB (server message block), caused worldwide disruption.

Bot and Botnets

Bot

- A bot is a software application that is programmed to do certain tasks.
- A bot is a software program that operates on the Internet and performs repetitive tasks.
- Although some bots are beneficial, malicious ones can severely damage a website or application.



Bot types (Good and bad bots)

1. **Chatbots**: Bots that simulate human conversation by responding to certain phrases with programmed responses. For example, **Apple's *Siri***, the virtual assistant pioneer, transforms the way users interact with their devices; **Alexa** from Amazon, **Google Assistant**.

2. **Web crawlers:** Bots that scan content on webpages all over the Internet. **Googlebots (Google search Engine crawler) - Googlebot Desktop and Googlebot Mobile**; **bingbo**t (Microsoft's search engine crawler for Bing), **Amazonbot** (the Amazon web crawler)

3. **Social bots**: Bots that operate on social media platforms. **Twitterbots** are already well-known examples. **Kismet** - a robot head that understands and exhibits emotion when interacting with humans.

4. **Malicious bots**: Bots that scrape content, spread spam content, or carry out credential stuffing attacks.

Malicious/bad bot activity

Any automated actions by a bot that violate a website owner's intentions, the site's Terms of Service. Malicious bot activity includes:

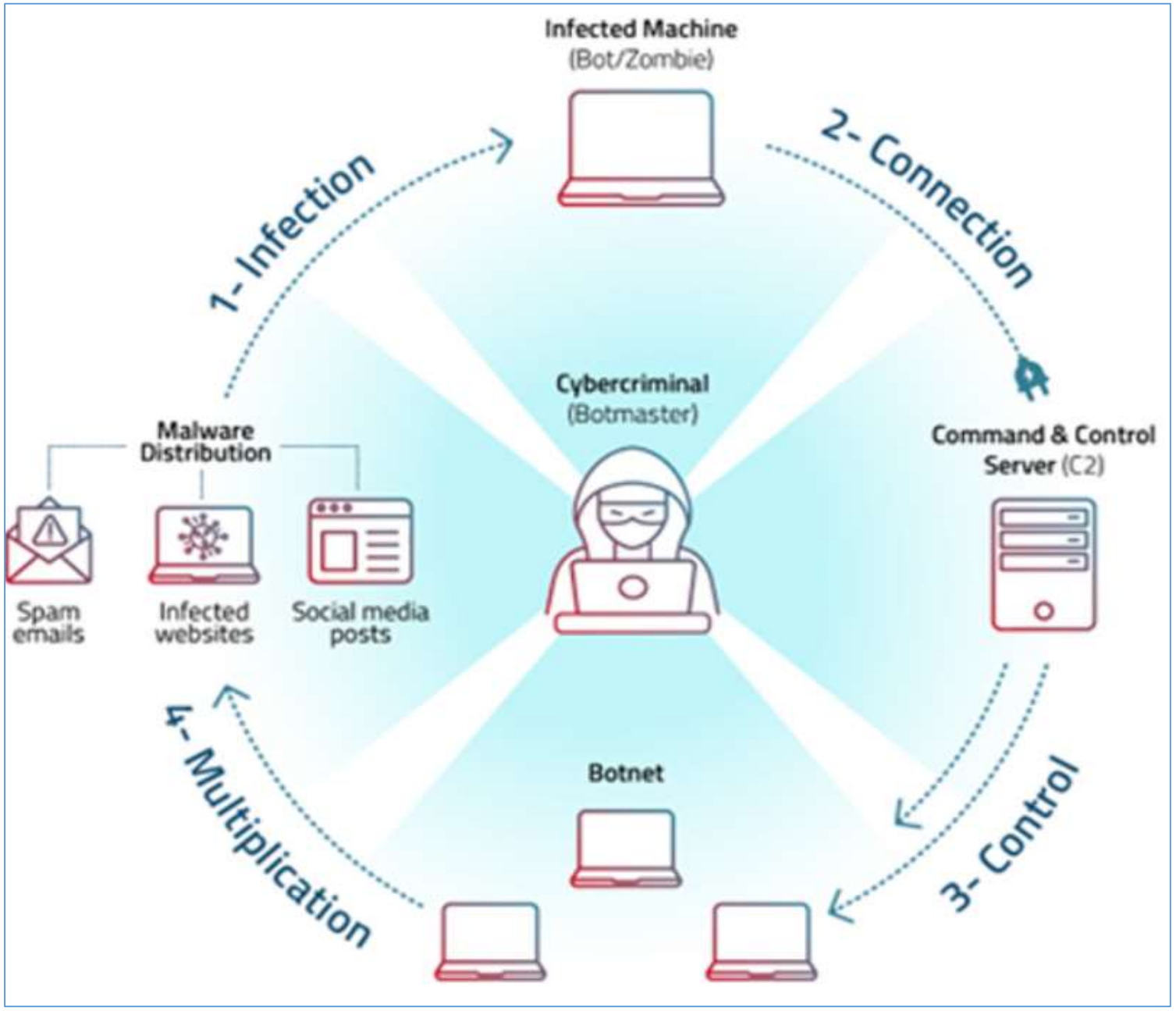
- Credential stuffing
- Web/content scraping
- DoS or DDoS attacks
- Brute force password cracking
- Inventory hoarding
- Spam content
- Email address harvesting
- Click fraud

Botnets

A botnet refers to a group of computers which have been infected by malware and have come under the control of a malicious actor called bot master.

or

A botnet is a network of internet-connected devices, like computers, smartphones, and IoT devices, that have been infected with malware and are controlled remotely by a cybercriminal called a "bot herder".



Privacy-invasive software

Privacy-invasive software: spyware and adware

Privacy-invasive software is a category of software that is built primarily to capture the target's private information. Basically, it ignores the privacy of the user and tracks the user activity. Some popular examples of privacy-invasive software are mentioned below:

1. Spyware

- It is a software that gathers the private information of an individual or an organization without their knowledge and pass this information to a third party. It is often used with a malicious intent. A few examples of **spyware are trojans, system monitors and tracking cookies.** These can collect any type of data from a system such as internet browsing habits of the users, bank account information, etc.

2. Adware

- Also known as advertising-supported software, this category is used for commercial intent**. It automatically displays unwanted** advertisements in order to generate revenue to its developer. The revenue is generated for two type of advertisements- one is for display and other is for pay-per-click.

Methods of Malware Attacks

Methods of Malware Attacks

The ways that malware can infiltrate and spread through networks continues to expand along with the threat vectors in the digital landscape, including the following:

- **Unsecure devices** that access the network such as personal mobile devices, PCs, and IoT devices open an attack vector for malware.
- **Unsecure networks** that are part of a supply chain or unsecure third-party partners' networks, can infect the networks of other suppliers or give malware access to the enterprise network.
- **Older devices** on the network with software that is not routinely updated can become compromised and spread malware.
- **Email attachments** containing malicious code can be opened and forwarded to other users, spreading the malware across the enterprise.

Methods of Malware Attacks

- **Phishing or spear phishing emails** trick the recipient into sharing passwords that give access to the corporate network where malware can spread.
- **Smishing texts**, similar to phishing emails but on mobile phones, trick distracted users into clicking on malware links and entering personal or business credentials that enable malware to spread on the network.
- **File servers**, such as those based on the common internet file system or network file system can spread malware as users download infected files.
- **File-sharing software** can allow malware to replicate itself onto removable media such as thumb drives and then on to computer systems and networks.
- **Peer to peer (P2P)** file sharing can introduce malware by sharing infected files as seemingly harmless as video, music, or images,
- **Remotely exploitable network vulnerabilities** can enable a hacker to access systems regardless of geographic location

Countermeasures against malware

Some countermeasures against malware

1. **Install antivirus software**: Install antivirus software on all devices connected to a network and update it regularly.
2. **Update software**: Keep your computer and software up to date with the latest versions.
3. **Use security tools**: Use a firewall, intrusion prevention system (IPS), and endpoint monitoring tools.
4. **Use email security**: Use email security filters to scan and remove malicious attachments and suspicious hyperlinks.
5. **Use web filters**: Use web filters to block access to websites that host malicious content.

Some countermeasures against malware

6. **Use multi-factor authentication (MFA)**: Use MFA to add an extra layer of security to your accounts.
7. **Back up your data**: Make multiple copies of your files using different backup solutions and storage locations.
8. **Use a non-administrator account**: Use a non-administrator account whenever possible. An **administrator account**, on the other hand, has full permissions to make system changes, including those that will affect other users.
9. **Develop security policies**: Develop security policies and implement security awareness training.
10. **Perform vulnerability assessments**: Perform routine vulnerability assessments

Economics of malware

Economics of malware

The economics of malware is driven by the desire of cybercriminals to make money, and the cost of cybercrime is expected to increase significantly in the coming years:

Malware costs

- The annual economic damage from malware worldwide is over $13 billion. Cybersecurity Ventures estimates that cybercrime costs will reach $10.5 trillion annually by 2025.

Ransomware

- Ransomware profits are expected to increase as criminals refine their techniques. For example, Check Point Research estimates that the Gandcrab ransomware attack generated $300–600,000 in ransom payments.

Malware distribution

- Malware can be installed when other programs are downloaded, including from third-party websites or peer-to-peer networks.

Every day, 560,000 new pieces of malware are detected. There are now over 1 billion malware programs in existence. Trojans account for 58% of all computer malware. Every minute, four companies fall victim to ransomware attacks.

Conclusion

- Malware stands as a persistent and evolving threat in the digital landscape, a shadowy counterpart to technological advancement.
- From simple viruses designed for mischief to sophisticated ransomware and state-sponsored attacks, its forms are as varied as its intentions are malicious.
- Combating this threat requires more than just robust antivirus software; it demands constant vigilance, proactive cybersecurity practices, and a collective responsibility from individuals, corporations, and governments alike.
- As our reliance on interconnected systems deepens, the battle against malware remains a critical and ongoing struggle to secure the integrity, privacy, and functionality of our digital world.